Combined Science: Electrochemistry

IGCSE Practice Paper

Total marks: 66 Estimated time: 79 minutes Questions: 22

Note: questions marked with * may stretch slightly beyond this chapter. They are included for stretch practice.

Write your answers in the boxes provided. Show working where appropriate.

Question 1. *[1 marks | Combined Science 2025 11]*

The diagram shows electrolysis apparatus. Which label shows the cathode?

- **A.** A (the battery/cell)
- **B.** B (the electrode connected to the negative terminal)
- **C.** C (the electrolyte)
- **D.** D (the electrode connected to the positive terminal)

Answer: _____

Question 2. *[1 marks | Combined Science 2025 12]*

Which statement describes what happens during reduction?

- **A.** An element gains oxygen.
- **B.** A compound gains oxygen.
- **C.** A compound loses oxygen.
- **D.** A mixture of elements loses oxygen.

Answer: _____

Question 3. *[1 marks | Combined Science 2025 11]*

Which statement explains why aluminium is extracted from its ore by electrolysis and not by heating with carbon?

- **A.** Aluminium is more reactive than carbon.
- **B.** Aluminium is more reactive than hydrogen.
- **C.** Aluminium melts at 660 °C.
- **D.** Aluminium resists corrosion.

Answer: _____

Question 4. *[1 marks | Combined Science 2025 13]*

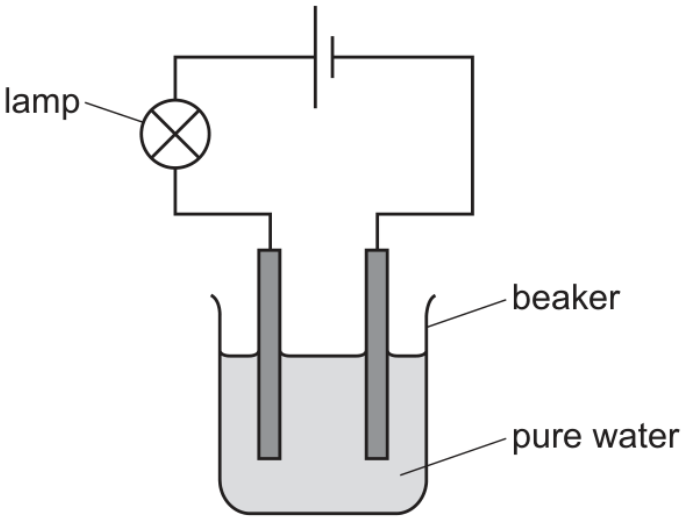
The diagram shows the apparatus for electrolysis with two electrodes labelled W (left, connected to the negative terminal) and X (right, connected to the positive terminal) dipping into a solution labelled Y. Which row identifies W, X and Y? (Columns: W, X, Y.)

- **A.** anode; cathode; electrolyte
- **B.** anode; cathode; power supply
- **C.** cathode; anode; electrolyte
- **D.** cathode; anode; power supply

Answer: _____

Question 5. *[1 marks | Combined Science 2025 22]*

19 The apparatus shown is used to test a property of compound R.



The lamp does **not** light when the beaker contains pure water.

When compound R is dissolved in the water, the lamp lights.

Which row describes compound R?

	type of bonding	elements of compound R
A	covalent	a metal and a non-metal
B	covalent	non-metals only
C	ionic	non-metals only
D	ionic	a metal and a non-metal

Question 6. *[1 marks | Combined Science 2025 12]*

18 What is produced at the anode during the electrolysis

A bromide ions

B bromine

C lead

D lead(II) ions

Question 7. *[1 marks | Combined Science 2025 23]*

Apparatus used to electrolyse molten lead(II) bromide is shown (d.c. power supply connected to two electrodes in molten lead(II) bromide; electrodes labelled P at top, Q at bottom/electrolyte). Which row identifies P and Q?

A. P = cathode; Q = electrolyte

B. P = anode; Q = electrode

C. P = anode; Q = electrolyte

D. P = cathode; Q = electrode

Answer: _____

Question 8. *[1 marks | Combined Science 2025 21]*

Which row shows what forms at the electrodes during electrolysis of an aqueous compound? Columns: anode, cathode.

A. anode metal, cathode non-metal

B. anode non-metal, cathode metal

C. anode hydrogen, cathode metal

D. anode hydrogen, cathode oxygen

Answer: _____

Question 9. *[1 marks | Combined Science 2025 22]*

27 Which row shows an electrolyte and the products of electrolysis for this electrolyte?

	electrolyte	products
A	concentrated aqueous sodium chloride	sodium and chlorine
B	molten potassium bromide	hydrogen and bromine
C	molten sodium chloride	sodium and chlorine
D	dilute sulfuric acid	hydrogen and sulfur dioxide

Question 10. *[1 marks | Combined Science 2025 12]*

22 Which statement describes the extraction of aluminium?

A It is extracted from bauxite using electrolysis.

B It is extracted from bauxite using carbon.

C It is extracted from hematite using electrolysis.

D It is extracted from hematite using carbon.

Question 11. *[1 marks | Combined Science 2025 21]*

27 An electric current is passed through molten calcium chloride using graphite electrodes.

Which row shows the electrode products?

	anode product	cathode product
A	calcium	chlorine
B	chlorine	calcium
C	chlorine	hydrogen
D	hydrogen	chlorine

Question 12. *[2 marks | Combined Science 2025 33]*

/ s	/ cm^3	/ cm^3
0	0	0
30	18	9
60	36	18
90	54	27
120		36

(i) Use Table 5.1 to predict the volume of gas given off at the negative electrode at 120 s.

volume = ... cm^3 [1]

(ii) Use Table 5.1 to calculate the volume of gas formed per second at the positive electrode.

volume = ... cm^3 / s [1]

[Total: 8]

Question 13. *[3 marks | Combined Science 2025 41]*

5 **(a)** Fig. 5.1 shows the electrolysis of molten zinc chloride, $ZnCl_2$, using inert electrodes.

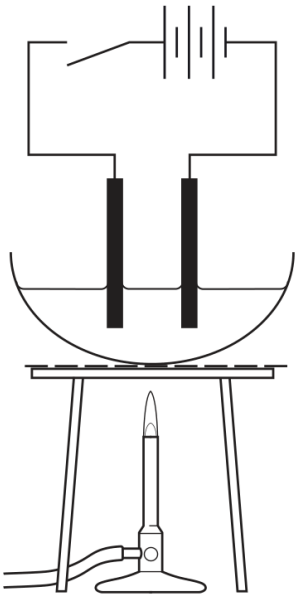


Fig. 5.1

(i) The switch is closed.

Describe the observation at the positive electrode.

.....

..... [1]

Question 14. *[3 marks | Combined Science 2025 32]*

(g) Draw **one** straight line from each description to the correct substance.

description	**substance**
	chlorine
salt that contains ionic bonds	
	dilute sulfuric acid
compound with a pH less than 7	hydrogen
	lead
Group VII element	
	lead(II) bromide

[3]

[Total: 11]

Question 15. *[5 marks | Combined Science 2025 32]*

1 **(a)** Fig. 1.1 shows a white blood cell.

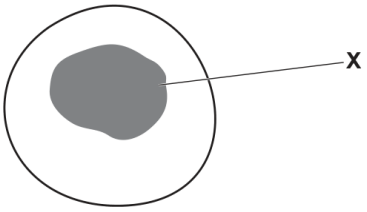


Fig. 1.1

(i) Name the part labelled **X** on Fig. 1.1.

..... [1]

(ii) State **one** function of white blood cells.

..... [1]

(iii) Circle **two** structures found in both white blood cells and bacterial cells.

cell membrane	**cell wall**	**cytoplasm**
circular DNA	**large vacuole**	**plasmid**

[2]

(b) Blood is part of the circulatory system.

Complete the sentence to describe the circulatory system.

The circulatory system is a system of blood ..... with a pump and ..... to ensure one–way flow of blood.

[2]

Question 16. *[5 marks | Combined Science 2025 43]*

4 Table 4.1 shows information about three compounds, **A**, **B** and **C**.

Table 4.1

compound	melting point / °C	boiling point / °C	electrical conductivity
A	801	1465	conducts only when molten or in aqueous solution
B	1713	2950	does **not** conduct
C	–210	–196	does **not** conduct

(a) **(i)** Identify the compound in Table 4.1 that is a solid at 25 °C **and** has only covalent bonding.

Explain your choice.

compound .....

explanation .....

.....

.....

[2]

(ii) Identify the compound in Table 4.1 that is a simple molecular compound.

Explain your choice.

compound .....

explanation .....

.....

.....

(b) The temperature of compound **A** is increased from 800 °C to 802 °C.

Describe how the arrangement, energy and motion of the particles in compound **A** change with this increase in temperature.

arrangement .....

.....

energy .....

.....

motion .....

.....
[3]

(c) An electric current is passed through an aqueous solution of compound **A**.

A chemical change occurs.

Explain why this is a chemical change.

.....

.....

..... [2]

[Total: 8]

Question 17. *[5 marks | Combined Science 2025 42]*

Continuation of Question 4 on sodium fluoride, now addressing the electrolysis of molten NaF (Fig. 4.2 shows inert electrodes dipped in molten sodium fluoride with a power supply) and the reaction of aqueous NaF with aqueous silver nitrate.

(c(i)) Define electrolysis. [2]

(c(ii)) Predict the product formed at the positive electrode and at the negative electrode during the electrolysis of molten sodium fluoride. [2]

(d) Aqueous sodium fluoride reacts with aqueous silver nitrate, AgNO3, to form aqueous sodium nitrate, NaNO3, and one other product. Write the balanced symbol equation for this reaction. [1]

Question 18. *[6 marks | Combined Science 2025 42]*

(c) The reaction between aluminium and iron(III) oxide is exothermic.

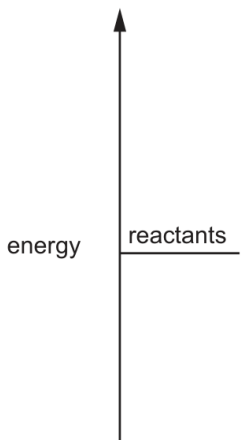
A flame is used to provide the activation energy to start the reaction.

Fig. 5.1 shows the incomplete reaction pathway diagram for this reaction.

Complete Fig. 5.1.

Include:

- the energy level of the products
- a labelled arrow to show the activation energy, E_a
- a labelled arrow to show the overall energy change of the reaction.



5 The reaction between aluminium and iron(III) oxide is used to join iron railway lines together.

$$2Al(s) + Fe_2O_3(s) \rightarrow 2Fe(l) + Al_2O_3(s)$$

Molten iron is produced in the reaction. Molten iron becomes solid when it cools.

(a) **(i)** State how the equation shows that **molten** iron is produced.

..... [1]

(ii) Use the equation to explain why this reaction is a redox reaction.

.....

.....

..... [2]

(iii) Complete the ionic equation for this reaction.

$Al + Fe^{3+} \rightarrow$ ..... + ... [2]

(b) Explain the meaning of the symbol (III) in iron(III) oxide.

.....

..... [1]

Question 19. ** [6 marks | Combined Science 2025 33]

Continuation of Question 4 on reaction between magnesium and steam.

(a(iii)) Complete the sentences about the reaction of magnesium with steam. Use words from the list (anions, cations, covalent, electrons, ionic, neutrons, neutralisation, oxidation, protons, reduction). Each word may be used once, more than once or not at all. During the reaction, magnesium atoms lose .... to form positive ions. These positive ions are known as .... . Oxygen atoms from the water molecules form negative ions. Positive ions and negative ions attract each other and form .... bonds. During the reaction, magnesium atoms gain oxygen. This type of reaction is .... . [4]

(b) The student repeats the experiment using copper instead of magnesium. Predict what the student observes during the reaction. Explain your answer. observation: .... explanation: .... [2]

Question 20. * *[6 marks | Combined Science 2025 31]*

(b) Fig. 5.2 shows part of the Periodic Table.

3 Li lithium 7	4 Be beryllium 9	5 B boron 11	6 C carbon 12	7 N nitrogen 14	8 O oxygen 16	9 F fluorine 19	10 Ne neon 20

Fig. 5.2

(i) Circle the word that describes the part of the Periodic Table shown in Fig. 5.2.

group **period** **property** **transition**

[1]

(ii) Identify an element in Fig. 5.2 that matches each description.

Each element may be used once, more than once or not at all.

- forms an ion with a charge of 1–

  .....

- has only two outer-shell electrons

  .....

- is used as an inert electrode in electrolysis

  .....

- reacts with oxygen to form a gas that leads to global warming

  .....

[4]

(iii) State how the metallic character of the elements shown in Fig. 5.2 changes from Li to Ne.

..... [1]

[Total: 8]

Question 21. *[6 marks | Combined Science 2025 33]*

5 Dilute sulfuric acid is used in an electrolysis experiment.

(a) Universal indicator is used to measure the pH of the dilute sulfuric acid.

Describe how to use universal indicator to measure pH.

.....

.....

.....

(b) Fig. 5.1 shows the electrolysis.

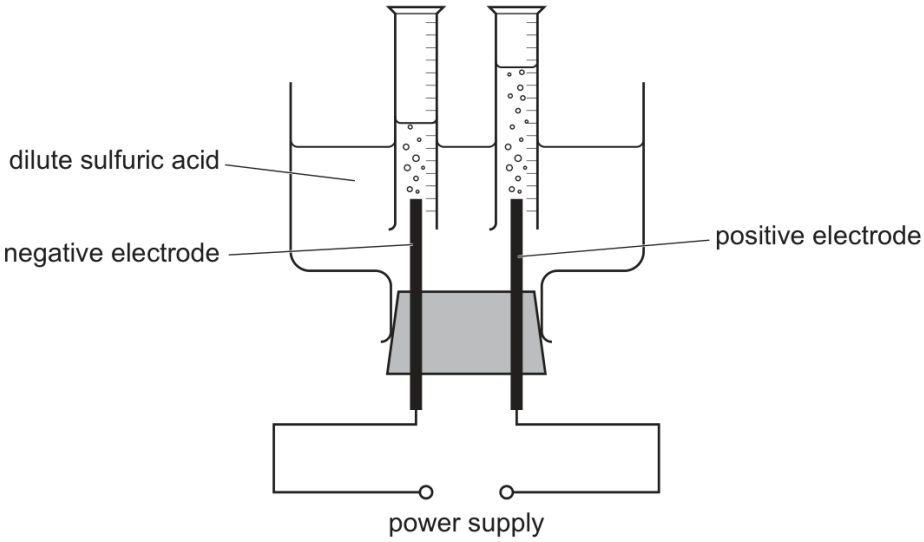


Fig. 5.1

(i) Identify the gas formed at each electrode.

negative electrode .....

positive electrode .....

(ii) State the name of the positive electrode.

.....

(iii) Identify the electrolyte in this electrolysis.

.....

(c) The volume of gas formed at each electrode is measured every 30 s for 2 min. The resu
shown in Table 5.1.

Table 5.1

time / s	volume of gas at the negative electrode / cm^3	volume of gas at the positive electrode / cm^3
0	0	0
30	18	9
60	36	18
90	54	27
120		36

(i) Use Table 5.1 to predict the volume of gas given off at the negative electrode at 1

volume = ... c

(ii) Use Table 5.1 to calculate the volume of gas formed per second at the positive ele

volume = ... cm

Question 22. *[8 marks | Combined Science 2025 32]*

6 Table 6.1 shows the products and observations at the electrodes in the electrolysis of three electrolytes.

Table 6.1

electrolyte	negative electrode		positive electrode	
	product	observation	product	observation
molten lead(II) bromide	lead	grey liquid	.....	bubbles of red-brown gas
concentrated aqueous sodium chloride	hydrogen	bubbles of colourless gas	chlorine	bubbles of green gas
dilute sulfuric acid	.....	bubbles of colourless gas	.....	bubbles of colourless gas

(a) Complete Table 6.1. [3]

(b) Explain why the mass of concentrated aqueous sodium chloride decreases during electrolysis.

.....

..... [1]

(c) Use Table 6.1 to explain why electrolysis is a chemical change.

.....

..... [1]

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